

# **SPRING BOOT SKILLS THAT INCREASE YOUR **SALARY****



@java\_in.depth

# JAVA 21 **VIRTUAL THREADS** + **SPRING BOOT 3**

**+15–25%** SALARY PREMIUM  
VS STANDARD SPRING MVC DEVS

Java 21's Project Loom introduced **virtual threads** — JVM-managed, not OS-managed. In Spring Boot 3.2+, one config line enables them: **spring.threads.virtual.enabled=true**. Result: your MVC app handles WebFlux-level concurrency without rewriting a single line to reactive code.

Most devs still run Java 11 or 17. **Knowing Java 21 VTs puts you in a rare bracket** — especially for high-traffic fintech and e-commerce systems.

Interviewers at Razorpay, PhonePe, Zerodha are asking about this in 2026. **Most candidates draw a blank.**

# RESILIENCE PATTERNS IN PRODUCTION

**₹12–18L** TYPICAL RANGE  
WHERE THIS GETS INTERVIEWED

Circuit Breakers, Retry, Rate Limiter, Bulkhead — via **Resilience4j**. Every microservice system fails. The question is whether your service fails gracefully or takes down the whole system with it. This is what senior Spring Boot devs know that mid-level devs don't.

**Circuit Breaker alone is asked in 60%+ of senior backend interviews.** Knowing the fallback pattern, half-open state, and thresholds is what gets you the offer.

Netflix OSS (Hystrix) is dead. **Resilience4j is the 2026 standard.** Devs who still reference Hystrix lose credibility instantly.

# OBSERVABILITY:

## MICROMETER + OPENTELEMETRY

**RARE SKILL.** < 15% OF SPRING DEVS  
HAVE THIS IN DEPTH • 2026

Distributed tracing across microservices using **Micrometer Tracing** (Spring Boot 3's replacement for Sleuth) + **OpenTelemetry**. Exporting traces to Jaeger or Zipkin. Custom metrics exposed via **Actuator + Prometheus + Grafana**. Knowing what's wrong in prod — not just that something is wrong.

**Spring Cloud Sleuth is deprecated.** Devs who still reference it in interviews signal they're behind. Micrometer is the current standard — and most devs don't know it.

MNCs and scale-ups hire specifically for this. "How do you debug a slow request across 5 microservices?" — this is your answer.

# EVENT-DRIVEN DESIGN

## WITH KAFKA + SPRING

**+30–50%**

SALARY PREMIUM

MICROSERVICES + KAFKA VS WITHOUT •  
GLASSDOOR 2026

Not just producing and consuming messages. **Exactly-once semantics, consumer group design, partition strategy, dead letter queues,** and handling rebalancing gracefully. Spring Kafka + Kafka Streams for real-time processing. This is the difference between "I've used Kafka" and "I've run Kafka in production."

UPI, Flipkart, Swiggy, PhonePe — **all run Kafka at their core.** Any fintech or e-commerce interview will probe async messaging. Most devs know the happy path. Know the failure scenarios.

**Transactional Outbox Pattern** — ensuring DB write and Kafka publish are atomic — is a 2026 senior interview question that eliminates 80% of candidates.

# SPRING SECURITY: BEYOND JWT BASICS

₹14–20L

ROLES THAT TEST THIS DEPTH  
FINTECH / BANKING / SAAS • 2026

Everyone knows JWT. What separates senior devs: **OAuth 2.0 Resource Server config, method-level security with SpEL, SecurityFilterChain customization** in Spring Boot 3 (**WebSecurityConfigurerAdapter** is gone — devs who use it fail the interview), and token introspection vs local JWT validation trade-offs.

**WebSecurityConfigurerAdapter** was deprecated in Spring 5.7, removed in Spring 6. Any dev still using it in 2026 is disqualified at BFSI and fintech companies immediately.

Multi-tenancy security, row-level security with Spring Data, and **integrating Keycloak or AWS Cognito** as the auth server — this is what senior devs ship.

# JVM PERFORMANCE TUNING + SPRING BOOT PROFILING

**TOP 10%**

SKILL RARITY AMONG  
SPRING BOOT DEVS IN INDIA • 2026

Reading a heap dump. Diagnosing a memory leak with VisualVM or async-profiler. Tuning G1GC vs ZGC for your workload. Spring Boot Actuator's /metrics and /health endpoints beyond defaults. Identifying N+1 query problems from slow query logs — before your manager does.

**Startup time optimization** using Spring Boot's lazy initialization, GraalVM native image compilation — shipping faster cold starts for Lambda/serverless is a 2026 demand at cloud-native orgs.

A dev who can find and fix a production performance issue is **10x more valuable than a dev who only ships features**. Senior interviews test this exact judgment.



# SPRING AI

## + AI-POWERED APIS

### EMERGING.

18% JOB LISTINGS SPIKE IN 2025  
NAUKRI • SPRING BOOT + AI ROLES

**Spring AI** (Spring's official AI abstraction framework) allows integrating OpenAI, Ollama, Anthropic, or Mistral into Spring Boot apps with familiar Spring idioms — ChatClient, EmbeddingModel, VectorStore. Building **RAG pipelines** with Spring AI + pgvector or Redis as vector store. This is where backend Java and AI intersect.

Product companies that ship AI features need Java devs who can **integrate LLMs without switching to Python**. Spring AI bridges this gap — and almost no one knows it yet.

**Early movers win.** In 2026, Spring AI knowledge on your resume opens interviews at scale-ups and MNCs that wouldn't have looked at your profile otherwise.



# THE SKILL GAP IS THE SALARY GAP.

- Virtual Threads (Java 21) — one config line, but most devs don't know why it works
- Resilience4j in depth — fallback, half-open state, bulkhead. Not just "I've used it"
- Micrometer + OpenTelemetry — Sleuth is dead. Stop referencing it in interviews
- Kafka: failure scenarios — Transactional Outbox, DLQ, exactly-once semantics
- Spring Security post-Boot 3 — SecurityFilterChain, not WebSecurityConfigurerAdapter